



Francisco Samagaio

Junior Data Engineer
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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Data Engineering | Python | SQL | ETL Pipelines | Data Lakes | Data Processing

Work Experience

Junior Data Analyst — Natixis in Portugal

- Developed and optimized SQL queries across ~300 financial reports in a large-scale Oracle data lake, improving query performance and ensuring data accuracy.
- Designed, maintained, and validated 30+ relational database tables supporting ETL pipelines and enterprise data workflows.
- Built and maintained automated data pipelines and reporting workflows using EasyMorph, supporting CIB analytics and operations.
- Led migration of ~80 reports from IBM Cognos to EasyMorph, rewriting and optimizing SQL logic for improved performance and system compatibility.
- Worked in a regulated Corporate & Investment Banking environment, supporting large-scale financial data processing, reporting, and data quality assurance.

Intern Developer — Natixis in Portugal

Dec 2023 – Dec 2024

- Developed Python-based automation pipelines replacing legacy PowerShell workflows, contributing to modernization of internal systems.
- Built ETL pipelines using Python (Pandas, NumPy) to process 150+ multi-format datasets (CSV, Excel, TXT, JSON).
- Automated multi-source data integration workflows, reducing manual processing effort and improving data pipeline efficiency.
- Worked with Git, Jenkins, and Linux environments to support CI/CD workflows and production systems.
- Supported batch data processing systems in a financial environment using COBOL, JCL, and CICS.

Research Intern / Master's Thesis — Fraunhofer Portugal AICOS

Feb 2023 – Jul 2023

- Developed a Master's thesis project involving generative AI applied to structured data generation for industrial design.
- Worked with Python, HPC environments, and Linux systems for large-scale experimentation and model training.
- Built and managed data processing workflows for model training and evaluation pipelines.
- Evaluated outputs using quantitative metrics and structured experimental pipelines.

Technical Skills

Programming: Python, SQL, C/C++

Data Engineering: ETL Pipelines, Data Modeling, Data Lakes, Data Integration, SQL Optimization

Data Processing: Pandas, NumPy, Batch Processing, Data Cleaning, Data Transformation

Infrastructure & Tools: Git, Linux, Jenkins, Jupyter, Bash

Databases: Oracle, Relational Databases

Education

Faculty of Engineering of the University of Porto — Master's Degree in Electrical and Computer Engineering
Master's Thesis: Generative AI for Ceramic Color Generation

2018 – 2023

University of Genoa — Erasmus+ Study Program

2022

Languages

Portuguese — Native; **English** — Fluent; **Spanish** — Intermediate; **French** — Beginner